EET202

HINES & TRANSFORMERS

MODULE:1

ure:2 – Armature Winding

edge about constructional details of DC machines

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Module – 1

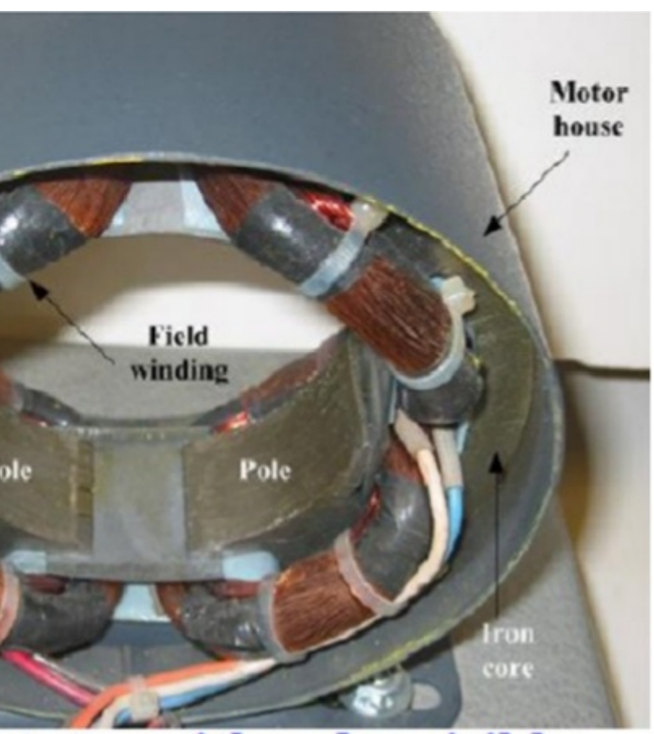
of dc machines - armature winding- single layer winding, double
wave, equalizer rings, dummy coils, MMF of a winding, EMF
etic torque - numerical problems.

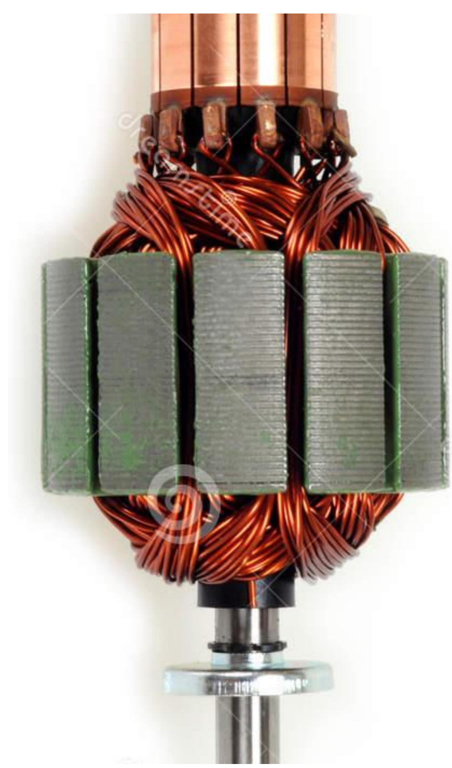
rmature windings

ngth of wire or strip in the slot which is responsible for
EMF.

ts of two conductors

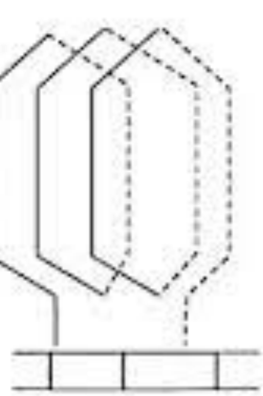
nsist of a single turn or many turns connected in series

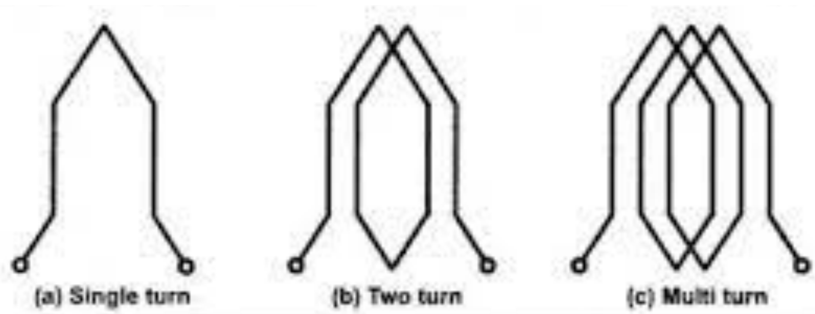




rmature windings

e length of wire or strip in the slot which is
he generation of EMF.
sists of two conductors
consist of a single turn or many turns connected in





tion of Armature windings

placement of conductors/coil sides in slots

ayer Winding

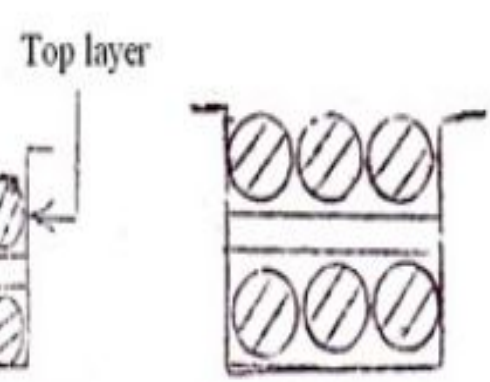
il side occupies the total slot area

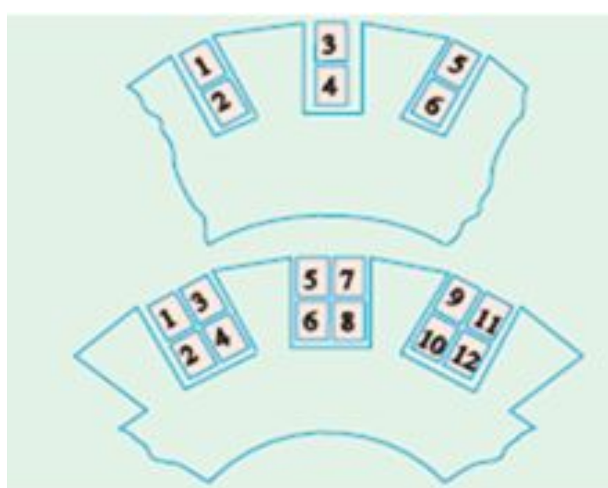
nly in small AC machines

Layer Winding

ntains even number of coil sides in two layers

chines generally use double layer winding



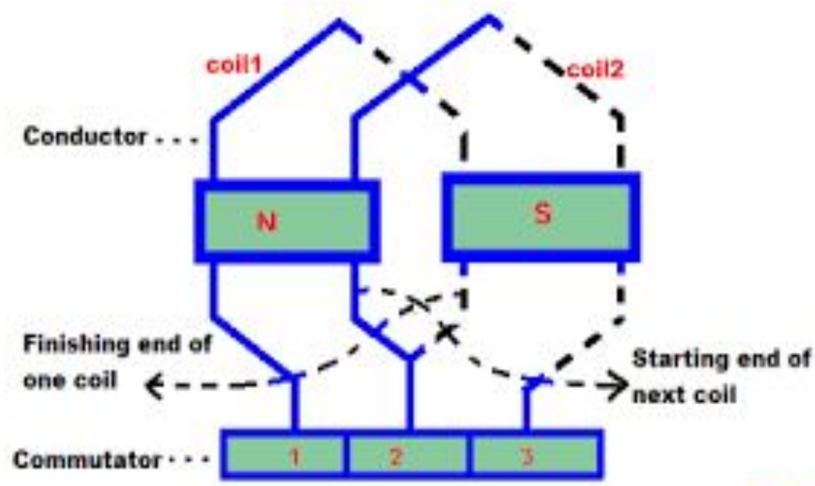


tion of Armature windings

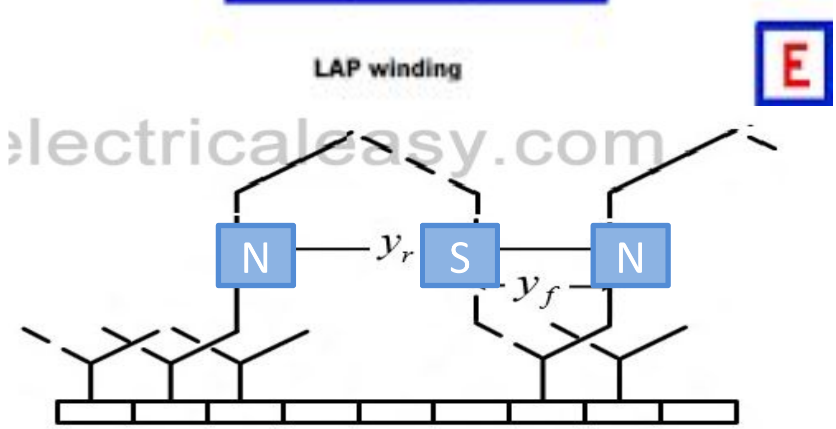
d connections

Finishing end of the
oe connected to the
f the next coil which
e same pole where
arted

: Finishing end of the
oe connected to the
f the next coil which
ne next similar pole
coil started



LAP winding

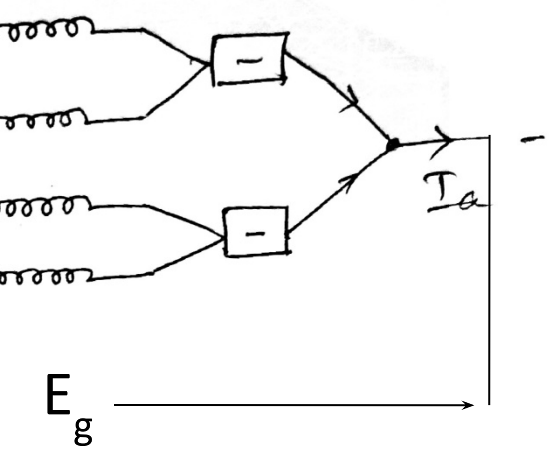


Wave winding

on – Lap & Wave Winding

DING	WAVE WINDING
e first coil can be tarting end of the ts from the same coil started. are divided into ns s = no. of poles es having large ge uired is equal to be required coils (50-500kW) and /) applications Current > 400A	• Finishing end of the first coil can be connected to the starting end of the next coil which starts from the next similar pole where the first coil started • Only two parallel paths independent of number of poles • Used for machines with low current and high voltage • Number of brushes required is always equal to two • No need of equalizer rings • Dummy coils may be required • Used for low power applications • Commonly used for currents < 400A

on – Lap & Wave Winding



n of Armature windings

tance between 2 adjacent main poles. It can be
erms of number of slots, circumferential length

h = 180 elec. Deg

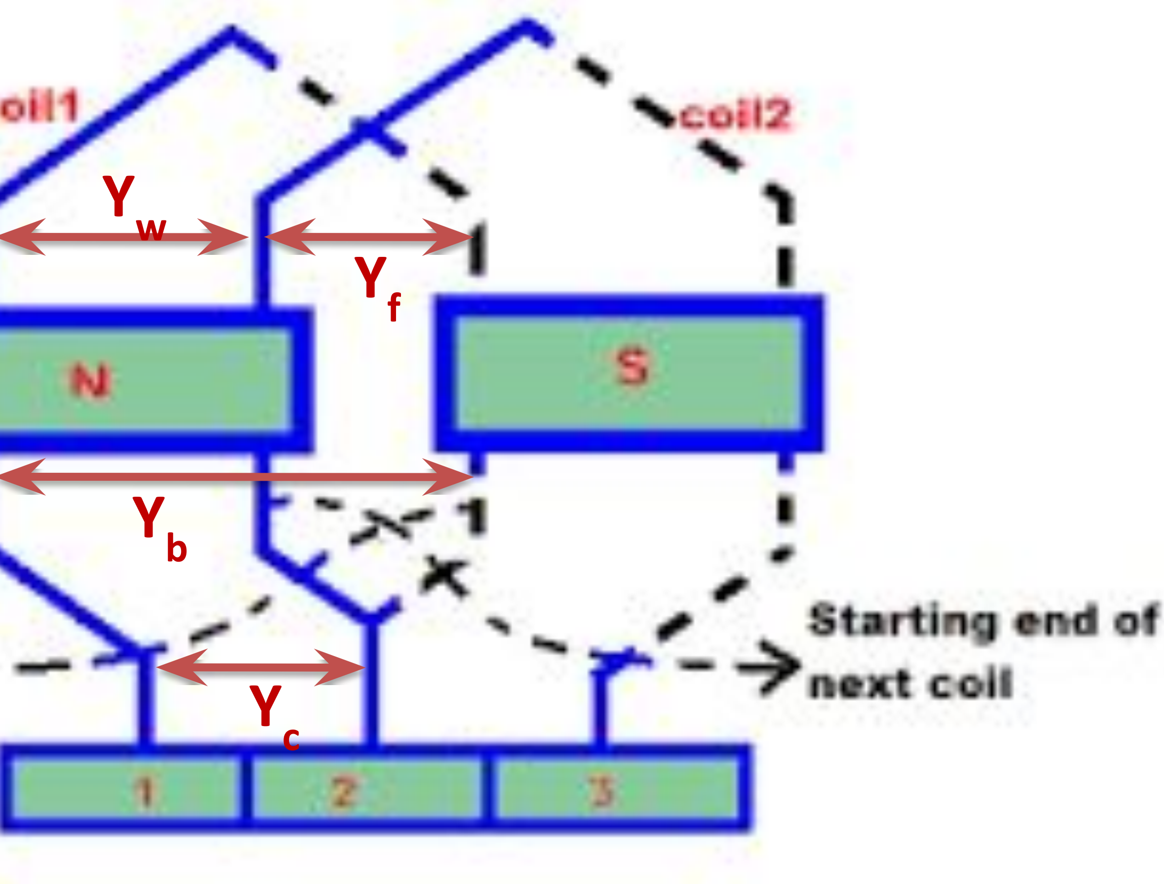
: Distance between two coil sides of the same

: Distance between 2 coil sides connected to
nutator segments.

or Resultant pitch(Y_w): Distance between
es of consecutive coils.

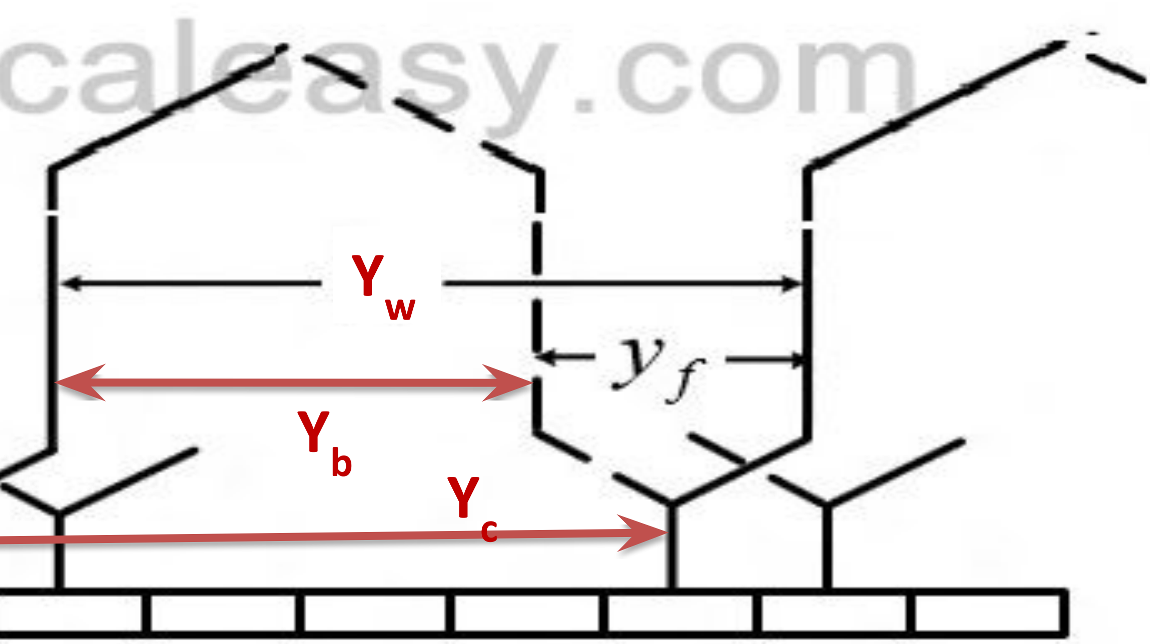
pitch(Y_c): Distance between 2 commutator
nich the 2 ends of one coil are connected.

n of Armature windings



LAP winding

n of Armature windings



Wave winding

n of Armature windings

Layer Simplex Lap Winding

$\frac{Z}{P} \pm K$ (where K is an integer or fraction to make back pitch an odd integer)

Z : Number of Conductor

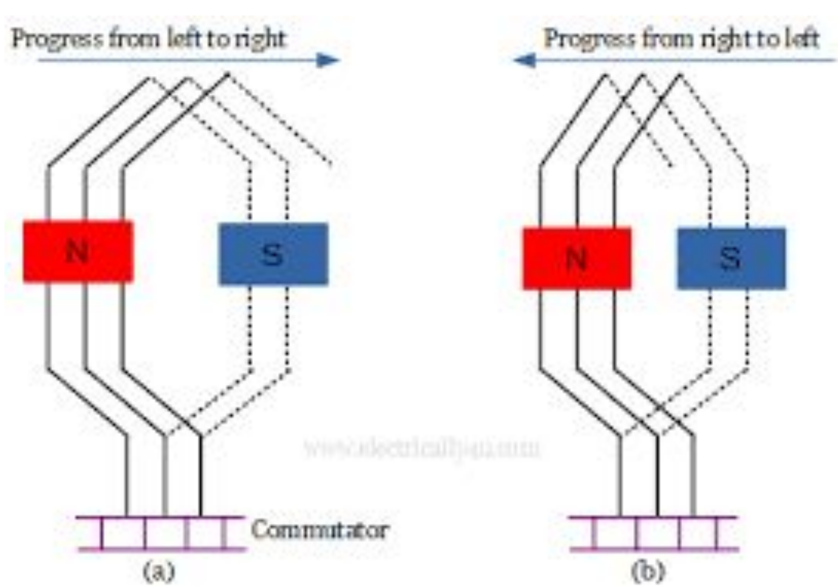
P : Number of Poles

$_w = \pm 2$

$= Y_b - Y_w$

winding

ve winding



n of Armature windings

Layer Wave Winding

$$Y_w = \frac{Z \pm 2}{P/2}$$

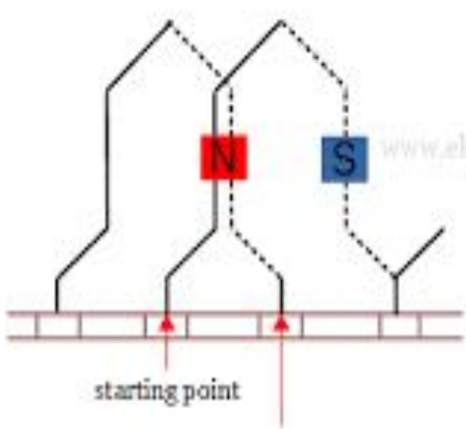
$_f$ should be odd)

onductor

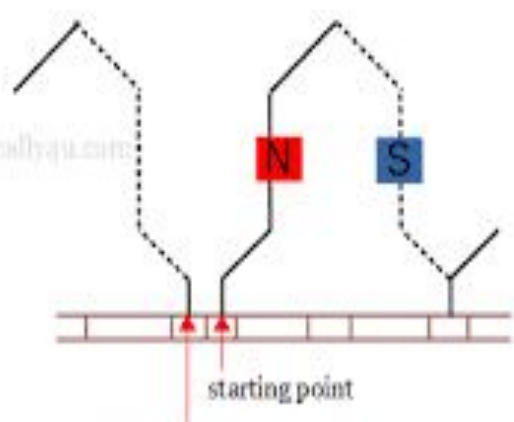
oles

ve winding

ssive winding



(a)



(b)

Problems

Simplex Lap Winding for a 4-pole, 14-slot, 2 coil-
c. armature, the back pitch y_b and front pitch y_f
ely

plex Lap Winding,

ors (Z) = number of coil sides

tch = Number of conductors / number of poles

t pitch should be odd and the difference between them is

s = 2 coil sides/slots

4

l sides = 2 × 14 = 28

ors (Z) = number of coil sides = 28

) = 4

28/4 = 7

2 = 5

Problems

pitch and back pitch for a 4 pole dc armature having onductors is provided with simplex lap winding.

Problems

tor with 16 coils has two layer lap winding. The

_______.

= 16

s two-layer winding

of conductors = 2 × 16 = 32

umber of conductors/number of poles = 32/4 = 8

Problems

itch and back pitch for a 4 pole, 13 slot, simplex wave erator with a commutator having 13 segments.

Problems

itch and back pitch for a 4 pole, 25 slot, simplex wave erator having 50 conductors.